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Voxelization based PoCA point cloud filtration algorithm for image reconstruction for Muon Tomography

R. Sehgal,[a,d,1] **M. Sengupta Mitra,**[b] **Tushar Roy,**[c] **S.T. Sehgal,**[a] **L.M. Pant**[a,d] **and B.K. Nayak**[a,d]

[a] *Nuclear Physics Division, Bhabha Atomic Research Centre,*
Trombay, Mumbai 400085, India
[b] *Health Physics Division, Bhabha Atomic Research Centre,*
Trombay, Mumbai 400085, India
[c] *Technical Physics Division, Bhabha Atomic Research Centre,*
Trombay, Mumbai 400085, India
[d] *Homi Bhabha National Institute,*
Anushakti Nagar, Trombay, Mumbai 400094, India

E-mail: rsehgal@barc.gov.in

Abstract: Multiple Coulomb Scattering (MCS) based muon tomography technique has been considered to be a well-known tool to identify, discriminate, and to image the high-density objects placed inside closed volumes. The two most famous reconstruction algorithms are Point of Closest Approach (PoCA) and Maximum Likelihood Expectation Maximization (MLEM). PoCA is fast but purely geometrical and as a result of this, it gives a lot of false positives, i.e. sometimes the PoCA point lies outside the target object and hence it forms an envelope of false-positive which results in a smeared image. On the other hand, MLEM is an iterative algorithm and is much more computation-intensive. In this work a new and innovative method is proposed which is based on the concept of voxelization to handle the known problem of false positives of the PoCA algorithm, and hence provide a clear reconstructed image. These algorithms remove the false positives PoCA points from the 3D point cloud and will give useful information in terms of regions or voxels within a voxelized volume 'V' to do a clear image reconstruction. The advantages of the proposed algorithm to the existing algorithms are also discussed. The status of the experimental setup of the proposed facility using Resistive Plate Chamber(RPC) with spatial resolution of $\sim$ 1cm as muon detector, is also discussed. The preliminary data from the current experimental setup, showing detector performance and cosmic muon tracks are also shown. Since the experimental setup is not fully ready, the effectiveness of the developed algorithms and the results are evaluated using the data from the Geant4 simulation of the muon tomography setup.

Keywords: Image filtering; Resistive-plate chambers; Simulation methods and programs

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[1] Corresponding author.

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1 Introduction

Muon tomography [1, 2] is considered as a promising technique for 3D imaging of Special Nuclear Materials (SNM), which needs discrimination between high Z and low Z materials. It uses the naturally available cosmic muons, which are continuously bombarding the earth with a limited flux of 1 muon cm^{-2} min^{-1} [3] at sea level. The muons are like the electrons but roughly 207 times heavier than them, having mean energy of 3–4 GeV at the earth's surface. Because of their high mass and

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energy, they can penetrate meters of high-Z or dense material. While passing through the material the muons deviate from their path as a result of Multiple Coulomb Scattering (MCS) and the angle of deviation depends on the Z as well as on the density of the material. Hence, cosmic muons can act as a probe to discriminate between high Z, low Z, and medium Z material [4–6]. These attractive properties make it a suitable candidate for homeland security applications like cargo scanning, to detect the illegal transport of special nuclear material which can't be done with traditional X-ray and gamma-ray based radiography. The scattering information along the muon trajectory can be measured by a pair of position-sensitive detectors. Various types of position-sensitive muon detectors exist like Gas Electron Multiplier (GEM) [7], Resistive Plate Chamber (RPC) [8, 9] and Drift-Tube detectors [10, 11], Drift Chambers [12]. The proposed muon tomography prototype setup make use of large area RPCs of 100 cm × 100 cm surface area, as muon detectors and eventually aims to discriminate and identify high Z material like Pb & U from medium and low Z material like Fe & Al and to do image reconstruction as well. However, the current work is mainly focused on the imaging part and proposes a voxelization based filtration algorithm that helps in the removal of false-positive PoCA points from the 3D PoCA point cloud to give a clear reconstructed image.

The organization of the paper is as follows: it starts with explaining the physics behind muon tomography, followed by the description of the experimental setup with RPC as a position-sensitive detector. Following this, the simulation of the muon tomography setup is described in detail, together with the reconstruction algorithm used. This is followed by the analysis of reconstruction algorithms based on the simulated data set. Finally, the image reconstruction and filtration results of the application of the proposed algorithm on simulated data are shown with objects of different shapes and materials.

1.1 Multiple scattering Muon Tomography

Muon Tomography works on the principle that the cosmic muon passing through the material is deflected by many small angles in the process of multiple scattering. The traversal is stochastic and it comes out of the material at an average scattering angle of θ. As a result of this scattering, it gets linearly displaced from unscattered exit point by distance x. The 2D projection of scattering angle follows Gaussian distribution with long tails, where the central 98% is well described by Gaussian with zero mean, whose standard deviation is given by Molier's equation (1.1)

$$\theta_0 = \frac{13.6MeV}{\beta c p}\sqrt{\frac{L}{X_0}}\left\{1 + 0.038\ln\left(\frac{L}{X_0}\right)\right\} \tag{1.1}$$

where p is the momentum of the muon, L is the thickness of the material, and X_0 is the radiation length [13] of the material and is given by equation (1.2)

$$X_0 \approx \frac{A \cdot 716.4\ g/cm^2}{\rho \cdot Z(Z+1)\ ln\ (287/\sqrt{Z})} \tag{1.2}$$

where A is the mass number, Z is the atomic number and ρ is the density.

2 Experimental setup

The experimental setup is developed as a prototype to detect and discriminate high Z material from low Z material. The setup consists of a cosmic Hodoscope, that is used to detect the traversal

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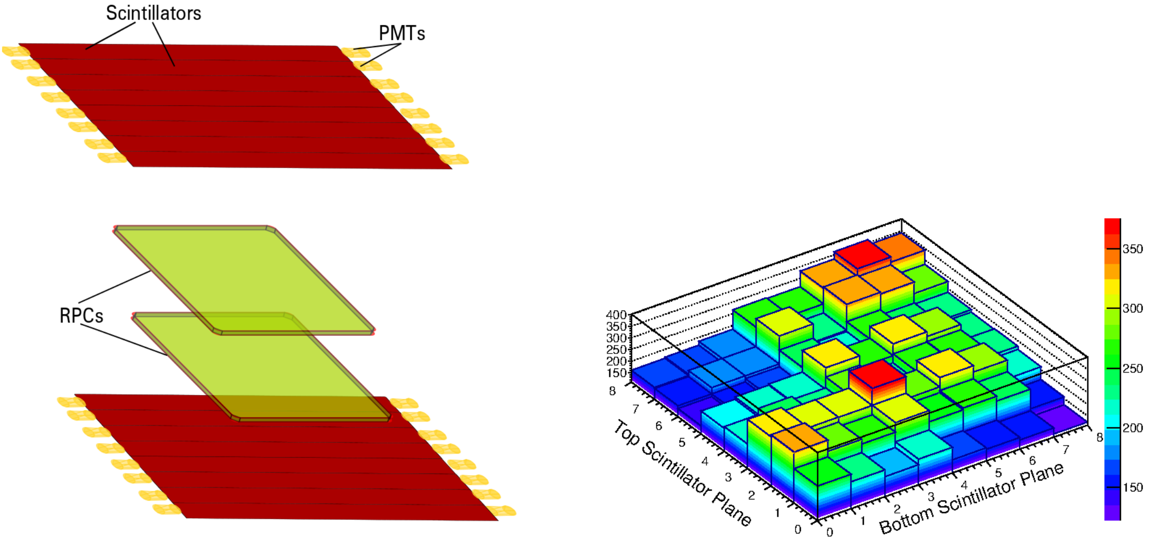
path of the particle in 3D space. The Plastic scintillator-based Hodoscope [14] was constructed at Nuclear Physics Division (NPD), Bhabha Atomic Research Centre (BARC) to characterize RPCs with cosmic muons, both for the RE4 (4th RPC Endcap disk) upgrade for the CMS experiment during the long shutdown (LS1) and also for the R & D related to glass RPCs for muon tomography and related experiments. The Hodoscope consists of two layers of large area scintillators that will be used to get cosmic muons triggers. Each layer consists of 8 such scintillators The dimension of each scintillator is 180 cm × 18 cm × 1 cm. As these scintillators are quite long, the Photomultiplier tubes (PMTs) are placed at each end, to ensure that sufficient scintillation photons reach at least one of the PMT and the efficiency of the scintillator is not underestimated. The OR of the signals from the two PMTs forms the scintillator signal. The OR of the eight signals from one scintillator plane forms the resultant signal for one layer. These two scintillator layers are separated by a vertical distance of 210 cm. When a muon passed through the top and bottom scintillator layer, the AND of signals from the two layers form the final trigger. The prototype setup uses RPCs as the position-sensitive detectors, which are placed between the top and bottom triggering layers as shown in figure 1a. Figure 1b shows the coincidence pattern between the scintillators of the top and bottom layers, using cosmic muons.



(a)

(b)

Figure 1. (a) Schematic showing RPC & Scintillator layout in the cosmic hodoscope. (b) Coincidence pattern of top and bottom scintillator planes from experimental setup.

2.1 Resistive Plate Chambers

The RPCs are the parallel plate gas detectors, built using the electrodes of high bulk resistivity such as glass or Bakelite. They have been used in the time of flight experiments for the detection of charged particle track, because of their excellent position and time resolutions. The RPCs that are used in the current work are glass RPCs, which are fabricated by assembling two glass plates having bulk resistivity of 10^{12} Ω-cm, separated by 2 mm gap using 2 mm thick spacer, forming a gas-gap. The gas-gap is filled with a gas mixture of Freon-R134a, Isobutane and SF_6 in the proportion of 95.2%, 4.5%, and 0.3% respectively. Freon serves as the medium for the interaction of incident

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particles. When muon interacts with Freon, ions and electrons are created. The recombination of these ions and electrons results in the generation of UV photons, which may result in a spurious pulse. To avoid this, Isobutane is used, that will absorb these photons, and hence, avoid the generation of the spurious pulse. SF_6 act as a quenching agent and helps in arresting the avalanche development. Each RPC contains two gas-gaps. Figure 2 shows the placement of these two gas gaps along with various other components inside the RPC. Two gas-gaps are used because the readout board used to fetch the signal from RPC responds only to negative signals, hence one gap is required for each of the X and Y planes. Readout planes used to read the signal have 32 strips each and are made from Cu clad G-10 sheets. Each of these 32 strips is 100 cm in length and 3 cm in width. The two readout planes are placed orthogonal to each other to confine the area of passage of muon and forms a grid of pixels of 3 cm x 3 cm dimension. When a muon passes through the RPC, it induces the signal in readout strips in the X and Y plane. These RPCs can operate either in an avalanche or in streamer mode. In the present work, the glass RPCs are used in avalanche mode. Two such RPCs are already assembled and data from these RPCs are analyzed to check their performance.

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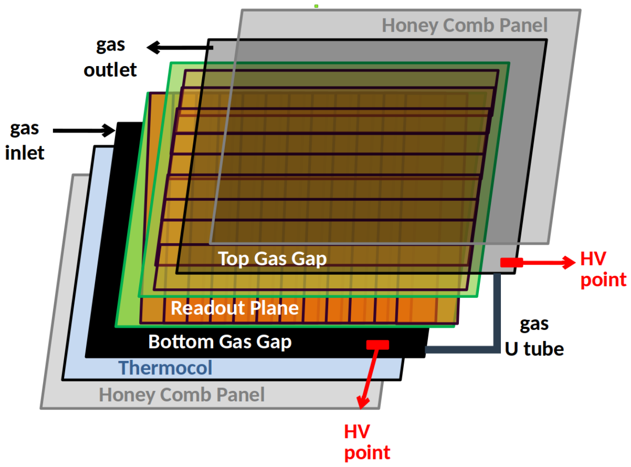

Figure 2. Schematic showing the layers of various components inside the RPC.

The general formula for the calculation of the spatial resolution of a detector with strip width of ‘w’can be given using

$$\sigma^2 = \mathbb{E}\left[x^2\right] - \left[\mathbb{E}\left[x\right]\right]^2 \tag{2.1}$$

The hit position within a pixel on the RPC can vary from $\frac{-w}{2}$ to $\frac{w}{2}$ with respect to center of the pixel. Considering the uniform hit distribution within the pixel

$$\sigma = \frac{w}{\sqrt{12}} \tag{2.2}$$

For the above mentioned RPC, w = 3 cm hence $\sigma = \frac{3}{\sqrt{12}}$ cm

2.1.1 I-V characteristics

I-V characteristics are done to check the quality of gas-gaps used in RPCs. The glass gas-gaps are first subjected to high voltage tests, where the applied high voltages are increased in steps and the leakage current is monitored. If the leakage current is higher than expected, the gap is rejected.

Figure 3 shows the variation of leakage current with an increase in voltage for each of the gas-gaps of one of the assembled RPC. The region from 0 to 8500 V is roughly linear and ohmic because the gas mixture shows high resistance and therefore leakage current is very small. At the voltage greater than 8500 V, the avalanche formation starts and the curve becomes nonlinear and this is the region where RPC needs to be operated in avalanche mode.

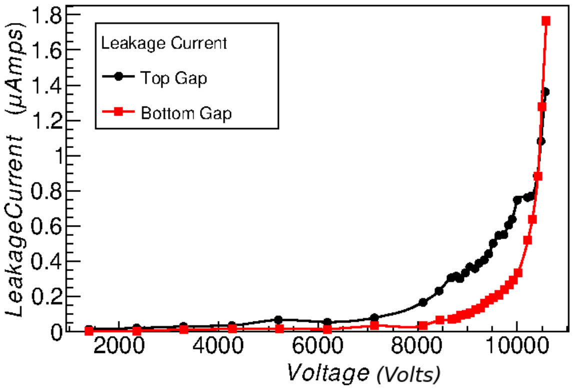

Figure 3. Variation of the leakage current with applied voltage across the top and bottom gas-gap.

2.2 Data Acquisition

The Data Acquisition system used in the setup is composed of VME and NIM standards. The VME controller used is CAEN Mod. V2718, which communicates with the PC via an optical link bridge. This VME controller can generate a clock of 20 MHz which is used to synchronize the TDCs. The TDCs used are CAEN Mod. V1190A and V1290A having 128 and 32 channels respectively. The intrinsic timing resolution of the TDC is 100 ps (with 40 MHz clock) and can accept either ECL or LVDS (low voltage differential signaling) signals only. The TDCs are being operated in the so-called ‘straddling’ mode [15] where the trigger window is straddled around the trigger. Straddling mode allows recording the events from the RPC detector, that occur both before and after the trigger, provided they are within the window width. The trigger matching mode can be set by tuning the different programmable parameters.

2.3 Preliminary data from experimental setup

Two glass RPCs with a surface area of 100 cm × 100 cm and separated by a vertical distance of 120 cm are placed in the above mentioned cosmic hodoscope. The center point of the pixel on the detector plane is considered as the muon hit point in the XY plane and the Z coordinate corresponds to the location of the detector in the vertical direction. With the current setup of two detectors, a track is constructed by making the straight line joining center point of pixel fired in each detector. For the setup with more than two detectors, a straight line fit will give a better estimate of hit point. The validity of the track is checked in the software by extended it to the top and bottom scintillator plane and determining the point of intersection and matched it with a scintillator strip that got fired for that event. If the scintillator strip matches then the track is valid and is kept to be used in the image reconstruction algorithm otherwise the track is declared as invalid and is removed. As the first step for track reconstruction with two RPCs, only those events are considered where only one strip fires in each plane in each detector. Figure 4 shows the tracks reconstructed using two RPCs and validated by matching the position with a fired scintillator strip.

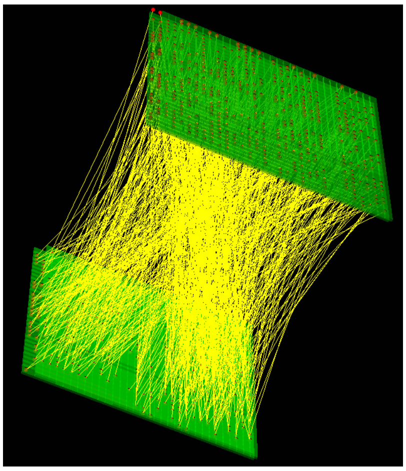

Figure 4. Tracks generated using experimental setup with two RPCs, considering center of the pixel as the muon hit position.

10000 events were reconstructed from the experimental setup with two RPCs. To evaluate the detector performance, the hit pattern of the individual strip in the X and Y plane, as well as the coincidence pattern of strips in X and Y planes, are plotted. Figures 5a and 5b show the strip profiles of two RPCs and figures 5c and 5d show the hit pattern of coincidence of strips in X and Y planes i.e. they show the hit pattern of the pixels of individual RPCs. These plots indicate that the full detector plane is firing. The variation of counts in pixels is expected to follow the geometrical acceptance of the detector but it is distorted by strip inefficiencies and noise.

3 Geant4 simulation

The simulations are done using Geant4 [16], which is a toolkit for the simulation of the passage of a particle through matter. The angular distribution and energy distribution of cosmic muons are sampled from the Cosmic-Ray shower generator (CRY) [17] library, which creates realistic cosmic particle distribution.

Figure 6 shows the cosmic muon angular distribution used in simulation. The angular distribution follow $\cos^2\theta$ [18] where θ zenith angle. To verify the $cos^2\theta$ distribution, fitting is done using the formula $mcos^n\theta$ in ROOT [19] and found the best fit value of fit parameters m = $2.445 \times 10^3 \pm 13.545$ and n = 2.025 ± 0.016.

The spectrum of the generated muons is established by the spectrum of cosmic ray muons in the range of 3 to 100 GeV, which obeys the empirical formula 3.1 [20]

$$\frac{dI}{dE \times d\cos\theta} = 0.014E^{-0.27}\left(\frac{1}{1+\frac{1.E\cos\theta}{115GeV}} + \frac{0.54}{1+\frac{1.1E\cos\theta}{850GeV}}\right) \tag{3.1}$$

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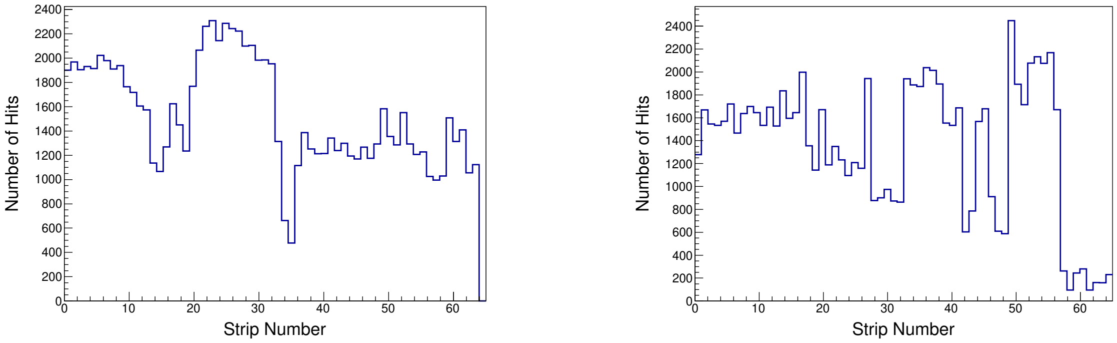
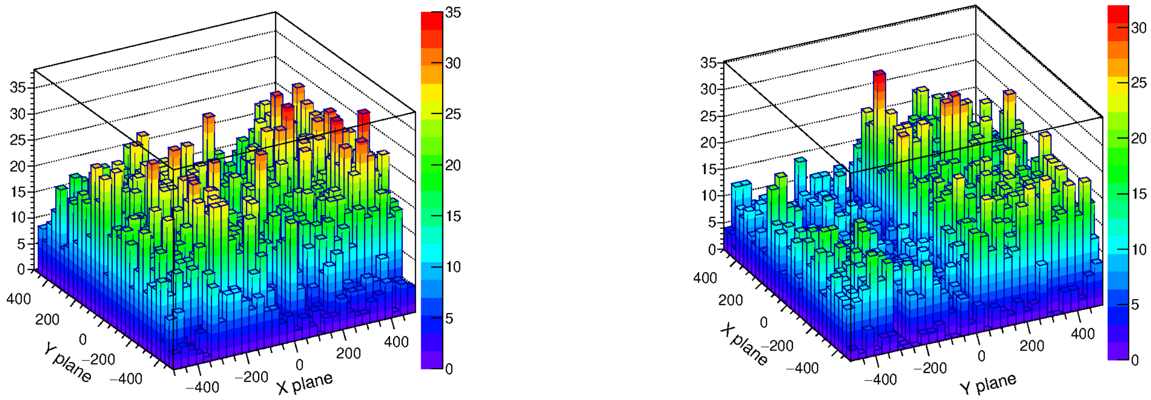


(a) Strip profile first RPC.

(b) Strip profile of second RPC.



(c) 3D Hit plot of the pixels of first RPC.

(d) 3D Hit plot of the pixels of second RPC.

Figure 5. (a) & (b) shows the strip profile of two RPCs. (c) & (d) shows the 3D hit plot of two RPCs kept at Z=+45 cm & Z=-45 cm.

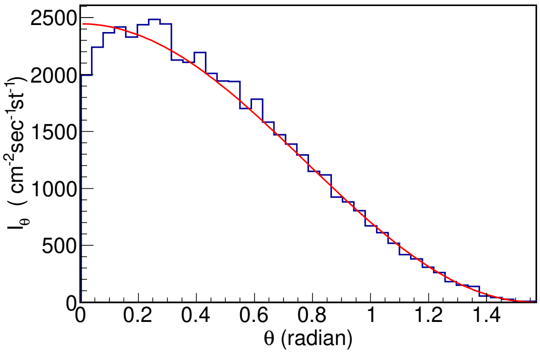


Figure 6. Angular distribution of CRY generated cosmic muons after applying solid angle correction and fitted using $mcos^n\theta$ with m = 2.445 $\times 10^3$ ± 13.545 and n = 2.025 ±0.016.

3.1 Simulation setup

The simulation setup consists of six resistive plate chamber detectors of 100 cm × 100 cm surface area. The vertical direction is considered as the Z-axis of the chosen coordinate system, and the detectors are placed symmetrically along the positive and negative Z-axis, with the origin at the

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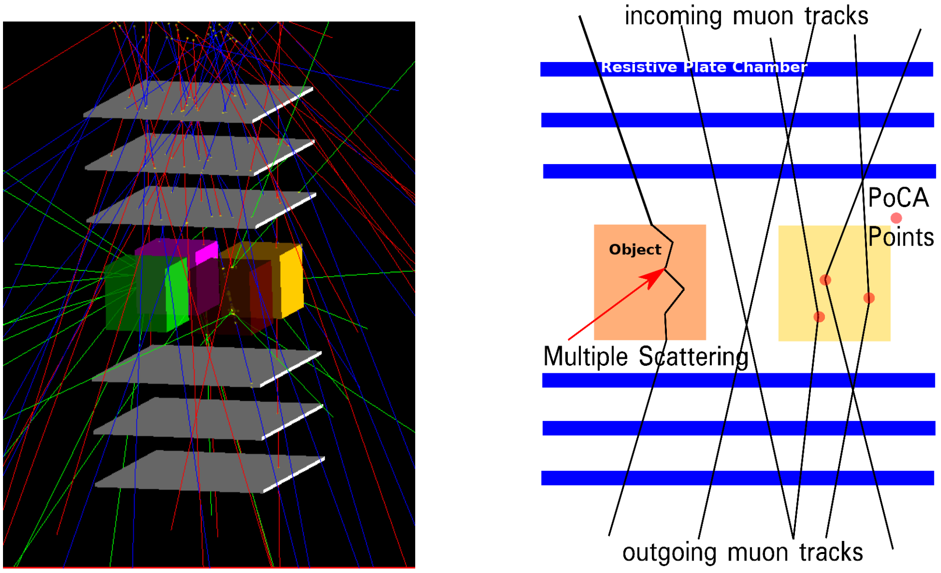
center of the setup. Only those muons are considered in the analysis which passes through all the six detectors. The top three detectors are placed at (0 cm, 0 cm, 105 cm), (0 cm, 0 cm, 75 cm), (0 cm, 0 cm, 45 cm) and the bottom three at (0 cm, 0 cm, -45 cm), (0 cm, 0 cm, -75 cm), (0 cm, 0 cm, -105 cm) respectively, and the scatterers are placed between the third and fourth detector as shown in figure 7a with air as the background volume. The chosen geometric setup is in accordance with the experimental setup described in section 2. In the simulation, the RPCs are considered to be 100% efficient , which implies that we will get the hit point from each detector. The top three detectors give us incoming muon direction and bottom three will give outgoing muon direction.



(a) (b)

Figure 7. Schematic showing (a) Geant4 simulation setup (b) Physics process under study, using the QGSP-BERT-HP physics list of Geant4.

The physical process of the muon, shown in figure 7b is described in QGSP-BERT-HP physics list of Geant4. The main physics process studied here is the Multiple Coulomb Scattering of cosmic muons when they pass through high Z material. It can be used to discriminate between the materials based on the fact that, the higher is the Z, the more is the scattering [21]. The objective here is to reconstruct the image of the object under test, using the incoming and outgoing muon tracks and the deviation in its path.

3.2 Track reconstruction algorithm

During the simulation, the track reconstruction study is done using the exact hit point from the detector as well as using the smeared hit point, where the degree of smear depends on the detector resolution. Track reconstruction using the exact hit point is done by making a straight line connecting the exact hit points. This is the ideal condition. But in reality, none of the detectors gives exact hit points. During the simulation, one needs to incorporate the detector resolution which in the current case is given by equation (2.2) where ‘w’ is 3 cm. Smeared hit point is generated on each detector plane by creating a uniformly distributed random point within $\pm\sigma$ along X and Y direction from the center of the pixel. A straight line fit to the smeared hit points gives the final track to be used by reconstruction algorithms.

4 Study of detector geometric acceptance

In the experimental setup, the detectors are placed along Z-axis and cover the XY plane only i.e. there is no detector coverage along with XZ and YZ plane. This will allow the triggering cosmic muon to pass through the volume under inspection without being detected. As a result, all these muons will get missed, which otherwise can be used for reconstruction. Also, as mentioned in section 3 that cosmic muon flux depends on the zenith angle and follows $cos^2\theta$ angular distribution. These two factors lead to a non-uniform detector geometric acceptance, which eventually increases the reconstruction time. The geometric acceptance study is carried out using the Geant4 simulation. The simulation setup consist top and bottom scintillator planes with cross-sectional area of 180 cm × 144 cm, placed at (0 cm, 0 cm, 105 cm) and (0 cm, 0 cm, -105 cm) respectively. The coincidence of these two scintillator planes is used as a trigger for the RPCs. Four RPCs of 100 cm × 100 cm cross-sectional area are placed between the two scintillator planes at (0 cm, 0 cm, 95 cm), (0 cm, 0 cm, 75 cm), (0 cm, 0 cm, -75 cm), (0 cm, 0 cm, -95 cm) respectively and the background volume is air. For the acceptance study, the volume between 2^{nd} and 3^{rd} RPC is the volume of interest, which is divided into 64, 000 voxels (2.5 cm × 2.5 cm × 2.2 cm). A voxelized volume navigator is implemented which keeps the record of all the muons that are detected by all the four detectors. The navigator finds all the voxels through which the muon has passed through and also counts the number of muons passing through individual voxel. The muon count of each voxel is normalized by the voxel with maximum count value. Finally, the normalized muon counts are histogrammed. A total 1 million muons are generated on the CRY plane of 2 m × 2 m area, placed at (0 cm, 0 cm, 125 cm) so that the generated muons cover the entire scintillator plane. This corresponds to 40 minutes of experimental data collection using cosmic muons. Out of 1 million generated muons, only $\sim$ 15% are detected by all the detectors. Figure 8 below shows the voxel slices at three different Z position (z = +42.9 cm, z = +1.1 cm, z = -42.9 cm) in the volume of interest. These slices show the normalized muon count that passes through the voxels at the corresponding Z value. The maximum acceptance is observed in the slices near the center of the volume as expected. Voxel slices that are far from the center along the Z direction are having less acceptance in both positive and in the negative Z direction. A similar pattern is observed along X and Y direction, where acceptance drop off quickly as we move away from the center along X direction or Y direction.

5 Image reconstruction algorithms

Image reconstruction algorithms for muon tomography depends on the measurement of incoming and outgoing muon tracks as they pass through the object under test. Various reconstruction algorithms [22–26] exist. The two most famous are the Point of Closest Approach (POCA) and Expectation-Maximization (EM) [27], which takes the incoming and outgoing track information and tries to reconstruct the 3D image. PoCA algorithm is described in detail in section 5.1. PoCA is a very fast algorithm and can quickly give the location of an object under test, but suffers from the problem of false positives. The proportion of false positives increases with coarse resolution

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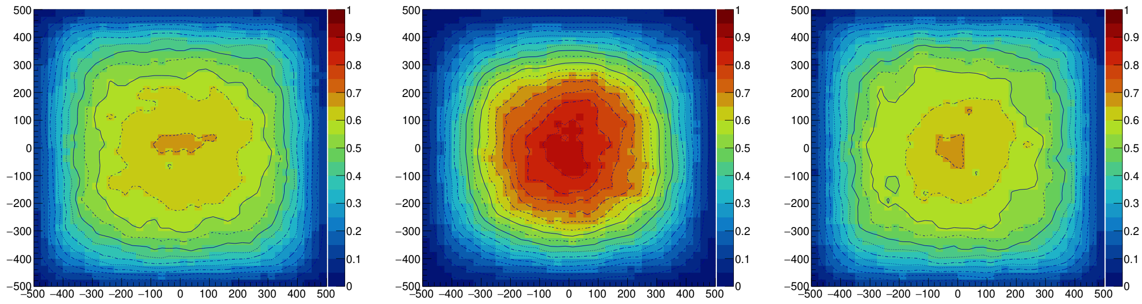

(a) Slice of Voxel at Z=+42.9 cm.
(b) Slice of Voxel at Z=+1.1 cm.
(c) Slice of Voxel at Z=-42.9 cm.

Figure 8. Relative acceptance of detector covering a volume of 100 cm × 100 cm × 90 cm using voxel of 2.5 cm × 2.5 cm × 2.2 cm. Three slices at different Z showing the variation of detector acceptance along Z.

detectors, like RPC in the current experimental setup. On the other hand, expectation-maximization based algorithms become slow due to their iterative nature. In this work, new filtration algorithms based on voxelization are proposed, that can quickly detect and remove the false positives PoCA points and give a clearer reconstructed image. The proposed algorithm also works well for the detectors with coarse resolution and tested on the data from simulation setup with RPC as position-sensitive detector. The filtered output can also be used as a good starting guess for likelihood-based iterative algorithm and may results in their fast convergence.

5.1 Point of Closest Approach (PoCA)

The Point of Closest Approach (PoCA) is a very simple and completely geometric algorithm. It uses the incoming and outgoing muon track and quickly calculates PoCA (the estimated point of interaction). Reconstruction of enough number of PoCA points results in the reconstructed image. The schematic of PoCA algorithm is shown in figure 9, where it tries to find the points $\mathbf{P_c}$ and $\mathbf{Q_c}$ on two skewed lines such that the distance between these lines is minimum, and PoCA point is the midpoint of the line joining $\mathbf{P_c}$ and $\mathbf{Q_c}$.

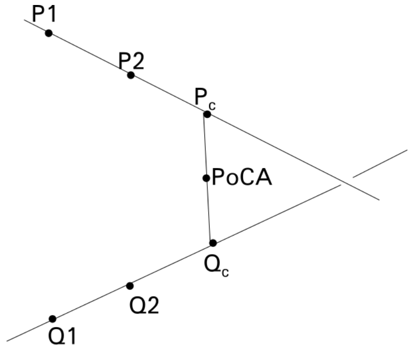

Figure 9. A schematic illustrating the working principle of the PoCA algorithm. P_c and Q_c are the points on the two skewed lines, where the distance between these lines is minimum. The mid point between P_c and Q_c is the Point of closest approach.

PoCA algorithm is very fast, and it is very good at providing the location of an object in 3D space. But it relies on single-point scattering and assumes that PoCA point is the point where single scattering happens and allocate the full scattering value to this point, which is not true in reality, because muon follows the stochastic path and suffer multiple scattering throughout the muon track. The PoCA method does not take multiple scattering into consideration. This assumption of single scattering results in a lot of false positives as shown in section 7. The analysis of the PoCA algorithm is done using simulated data, to see the proportion of false positives and true positives by calculating various performance metrics shown in section 6.

5.2 POCA point cloud filtration

The problem of removal of the false-positive PoCA points is targeted here. As mentioned in section 1, that distribution of angular deviation of muon during passage through the material follows a Gaussian distribution. The algorithm takes advantage of this and works on the basis that the PoCA points from the muon tracks, passing through the material will form a dense cluster in 3D space. The false-positive points form an envelope around the cluster and result in a smeared reconstructed image. An outlier detection algorithm can be used to identify and remove the false positive envelope to get a clear reconstructed image. To achieve this, various unsupervised clustering, and outlier detection algorithms like K-means [28], DBSCAN [29] are explored. Outlier detection using K-means clustering is done using a cluster-based approach where points in the smallest cluster are considered as an outlier and hence, removed from the data. K-means divides the points in 3D space into ‘K’ number of clusters by minimizing the sum of squares within the cluster, where the value of ‘K’ is the input to the algorithm. DBSCAN belongs to the class of algorithm that finds clusters using a distance-based approach. It is a density-based spatial clustering algorithm. To find clusters it needs the following pieces of information, **eps** (the neighboring distance for the point to be a part of a cluster), **minPoints** (the minimum number of points to declare the cluster a signal and not the outlier). It’s better than K-means because it does not require the user to supply the number of clusters that one wants to identify. These unsupervised clustering and outlier detection algorithms may be considered as a possible filtration method, but they are not deeply investigated in this work as they required a certain number of parameters to be tuned, which are not available directly from the point cloud of tomography data. To overcome these difficulties, voxelization based algorithms are proposed which are completely parameter-free and uses the concept of voxelization to discretize the 3D volume for further processing of 3D point cloud. In this work, two voxelization based filtration algorithms are proposed and analyzed. The first one is based on the application of a threshold value to the PoCA point count in a voxel and is very effective for the detector system with very fine resolution. The second one is based on the application of weight factor to the PoCA point count in a voxel and is useful for the detector system with coarse resolution, like RPC in the current experimental setup.

5.2.1 Voxelization

Voxelization is concerned with converting the geometric object or region from their continuous geometric representation into a set of small voxels that best approximates the continuous objects. It is widely used in computer graphics as well as in various physics simulation software, where it is used to do fast and efficient particle navigation. In Muon tomography, it is used to inspect the small region of volume and to find various physical parameters (radiation length, scattering density, etc.) associated with the volume of a voxel. In this work, voxelization is used to filter out the false-positive POCA points. Figure 10a shows the test simulation setup with six RPCs placed at (0 cm, 0 cm, 105 cm), (0 cm, 0 cm, 75 cm), (0 cm, 0 cm, 45 cm), (0 cm, 0 cm, -45 cm), (0 cm, 0 cm, -75 cm) and (0 cm, 0 cm, -105 cm) respectively. Four cubical Pb scatterers of 20 cm × 20 cm × 20 cm dimension, are placed between the third and fourth detector.

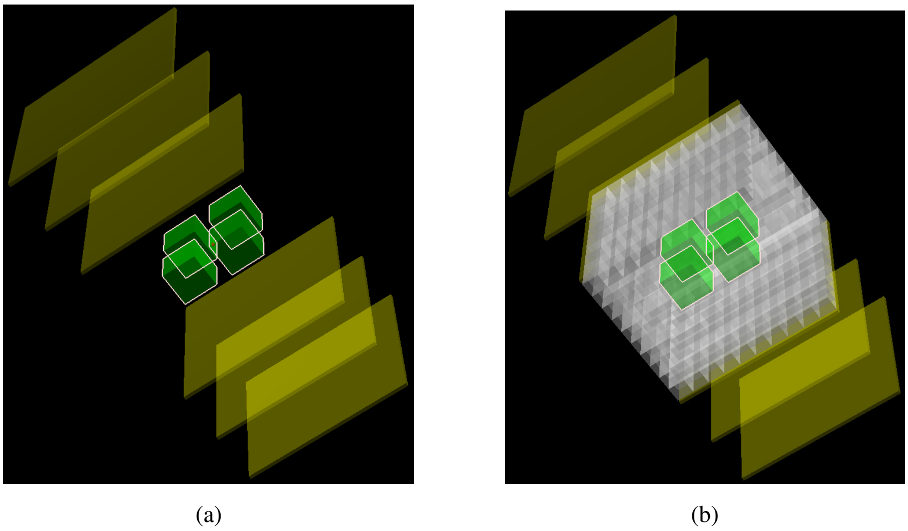
(a) (b)

Figure 10. (a) Simulation setup with six RPCs and four cubical scatterers of lead placed between the middle two detectors (b) Same simulation setup, but the region between middle two detectors is voxelized into 1000 voxels.

To apply voxelization, the region between middle two detectors (100 cm x 100 cm x 90 cm) is divided into N(= $N_x \times N_y \times N_z$) 3D voxels, where N_x, N_y and N_z are the number of divisions along X, Y and Z direction, in which the region between the middle two detectors are divided. To make the picture clearer, figure 10b shows the setup with 1000 voxels, i.e the length of the voxelized volume is divided into 10 equal parts in each direction. The indexing of these voxels is equivalent to the indexing scheme used for the 3D array. The voxelized volume is divided into N_z layers along Z direction, and each layer contains $N_x \times N_y$ voxels. For each of the incoming muon, the PoCA point is calculated and inserted in the voxel which corresponds to the geometric location of the PoCA point. Once all the PoCA points are processed, thresholding criteria is applied to remove the false positives.

5.2.2 PoCA count thresholding based point cloud filtration

As mentioned above, the PoCA algorithm introduces a lot of false positives. An algorithm is required to remove the false positives envelope and to give a clearer reconstructed image. To counter this problem, a statistical thresholding algorithm is proposed, where the threshold value defines the minimum number of PoCA points that a voxel must contain, to be declared as a signal voxel. Threshold selection is done by considering the outermost envelope of voxels of the volume of interest. This envelope may be considered as an offset region of the volume to be inspected, i.e. material will never be placed in this region. This implies that all the PoCA points that lie within this region will be false positives only. The histogram of the PoCA point count of this voxel envelop is used to calculate the threshold value using a mean + 3σ range. If the PoCA point count of a voxel in the remaining region is less than this threshold value, the voxel will be considered as a

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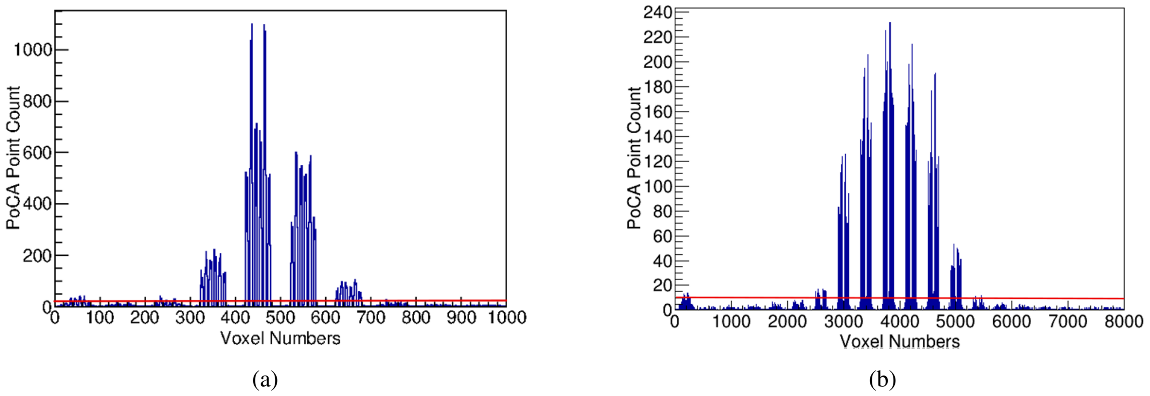
noisy voxel and all the PoCA points belonging to that voxel will be removed from the reconstructed image, otherwise, it will be considered as a signal voxel.



(a) (b)

Figure 11. (a) Distribution of PoCA point counts in (a) 1000 voxels and (b) 8000 voxels and the calculated threshold for ideal simulated data using PoCA point counts of the envelope of outermost voxels.

Figure 11a and 11b show the distribution of PoCA point counts in 1K and 8K voxels respectively. The red line shows the calculated threshold value using the envelope of outermost voxels, and clearly, this can remove most of the noisy voxels (smaller spikes). The bin number of bigger spikes corresponds to the voxel numbers where the material is placed.

Algorithm 1: pseudo code for PoCA point count thresholding based filtration algorithm.

@Input

1) PoCA point cloud
2) Threshold value
3) Total Number of Voxels

@Output

Filtered and Correlated PoCA point cloud

```
$T \leftarrow ThresholdValue$
        (based on the outermost envelope of Voxels )
$N \leftarrow Total\ Number\ of\ Voxels$
for $i \leftarrow 1\ \ to\ \ N$ do
    $C_i \leftarrow Number\ of\ PoCA\ Points\ in\ Voxel_i$
    if $C_i < T$ then
        $Remove\ all\ the\ PoCA\ Points\ belongs\ to\ Voxel_i$
        (considered as false positive PoCA points)
    else
        $Keep\ all\ the\ PoCA\ Points\ belongs\ to\ Voxel_i$
        (considered as true positive PoCA points)
return
```

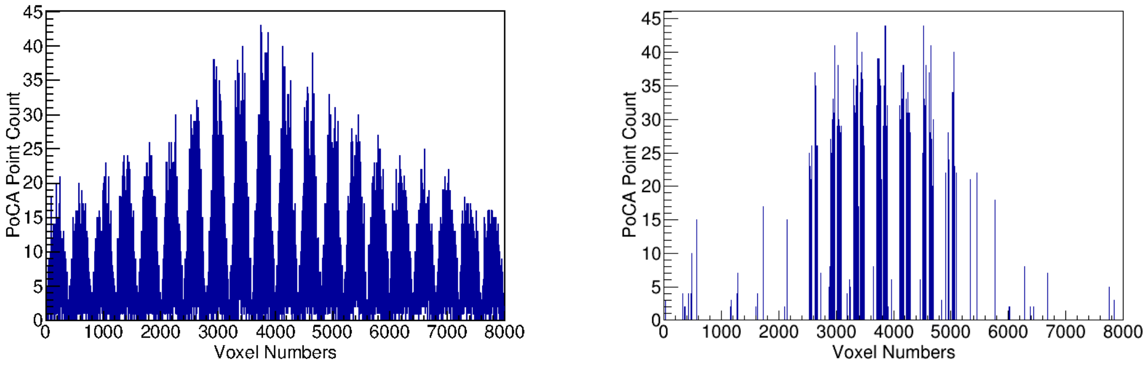
5.2.3 Point cloud filtration using standard deviation of scattering angle of PoCA points in a voxel as weighting factor

The above-mentioned algorithm works pretty well for ideal data but does not work well with the smeared data, where the detector does not have a very fine resolution. The resolution of RPCs used in the experimental setup is given by equation (2.2). The uncertainty in the estimation of muon hit position due to detector resolution results in an exponential increase in the number of false positives, which is evident from figure 12a, where now there are bigger spikes in the entire range of voxels, hence, it is difficult to choose a threshold value that will remove the false positive PoCA points. To handle the problem of false-positive due to smeared data, another algorithm is proposed, which works at finer level i.e. at the level of an individual voxel.

It is observed that the standard deviation of the scattering angle of PoCA points in the voxels containing the false positives is much less than compared to the voxels where the material is placed. Based on this observation, the standard deviation of a voxel is chosen as the weighting factor for its PoCA point count, i.e. PoCA point count in the voxels are replaced with weighted PoCA point count as mentioned in the equation (5.1).

$$Weighted\ PoCA\ point\ count = \sigma_{\mathrm{voxel}} \times PoCA\ point\ \ count\ in\ a\ voxel \tag{5.1}$$

Figure 12b shows the distribution of PoCA point counts in 8K voxels after application of standard deviation of a voxel as weighting factor for its PoCA point count. The threshold value based on the outer envelope of voxels is now calculated using weighted PoCA point count. If the weighted count in a voxel is greater than the calculated threshold value, the voxel is considered as a signal voxel, otherwise, it is considered as a noisy voxel, and all the PoCA points belong to that voxel are removed from the reconstructed PoCA point set.



(a)

(b)

Figure 12. (a) Distribution of PoCA point counts in 8K voxels for smeared data. (b) Distribution of PoCA point count in 8K voxels after application of standard deviation as a weighting factor to PoCA point count.

Algorithm 2: pseudo code for Weighted PoCA point count thresholding based filtration algorithm.

@Input
1) PoCA point cloud
2) Total Number of Voxels

@Output
Filtered and Correlated PoCA point cloud from smeared data

```
$N \leftarrow Total\ Number\ of\ Voxels$
for $i \leftarrow 1\ \ to\ \ N$ do
    $C_i \leftarrow Number\ of\ PoCA\ Points\ in\ Voxel_i$
    $SD_i \leftarrow Standard\ deviation\ of\ scattering\ angle\ of\ PoCA\ Points\ in\ Voxel_i$
    $Weighted\ Count\ in\ Voxel_i\ (WC_i) \leftarrow C_i \times SD_i$
$T \leftarrow ThresholdValue$
(based on the outermost envelope of Voxels where $Voxel_k$ now contains $WC_k$ PoCA points.)
for $i \leftarrow 1\ \ to\ \ N$ do
    if $WC_i < T$ then
        $remove\ all\ the\ PoCA\ points\ belongs\ to\ Voxel_i$
        (considered as false positive PoCA points)
    else
        $keep\ all\ the\ PoCA\ points\ belongs\ to\ Voxel_i$
        (considered as true positive PoCA points)
return
```

6 Algorithm's performance analysis

To do the performance analysis of PoCA and for the above-mentioned filtration algorithms, simulated data is generated for the setups as shown in figure 10a. The simulation ran for 10^5 reconstructed events, where the muons pass through all the 6 RPCs of 100 cm × 100 cm surface area. With the assumption (as mentioned in section 3.1), that the detectors are 100% efficient , one will get 10^5 tracks which pass through and are detected by all the detectors in the setup. To evaluate the performance of the algorithms, PoCA Accuracy Ratio (PAR) and True Positive Detection Ratio (TPDR) are calculated which are described in detail below:

6.1 PoCA Accuracy Ratio

The PoCA Accuracy Ratio (PAR) defines the degree of true or false positives PoCA points under a completely known information setup, which includes known location of scatterers and also the information that whether the muon track hits the scatterer or not. It is a useful parameter to evaluate an algorithm on the simulated dataset, where all this information is available. Ideally, if the track hits the scatterer then the interaction points must lie inside the scatterer region, but if it lies outside, then it represents false-positive, which will eventually form a noisy point cloud around the scatterer region and will create a smeared reconstructed image. Mathematically it is defined as the ratio of the number of PoCA points that lie within the region of scatterer divided by the actual number of muon track that hits the scatterer.

$$PAR = \frac{Num.\ of\ PoCA\ points\ that\ lie\ within\ the\ scatterer}{Num.\ of\ muon\ tracks\ that\ really\ hits\ the\ scatterer}$$

To study the PoCA accuracy ratio, simulation is done with four cubical Lead blocks of 20 cm × 20 cm × 20 cm dimension as scatterer, placed at (-15 cm, -15 cm, 0), (-15 cm, 15 cm, 0), (15 cm, -15 cm, 0), (15 cm, 15 cm, 0) locations. The schematic is shown in figure 10a and the corresponding PAR graph is shown in figure 13. From here one can see that even if we consider only those tracks that hit the scatterer, still, there are around 18% false positive that will form the envelope around the object, and hence, introduce the smear in the reconstructed image. As mentioned in section 5.1, the PoCA algorithm can quickly tell the location of the object under test, provided we get the clear reconstructed signal. Hence, we need to get rid of false positives upto an extent where we get a clear reconstructed image of the scanned object.

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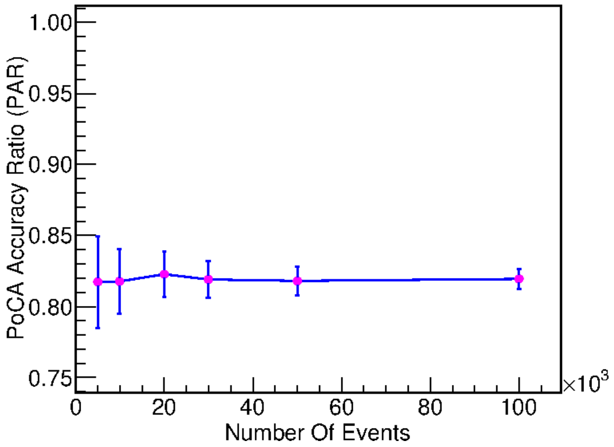

Figure 13. Variation of PAR with number of events.

6.2 True Positive Detection Ratio (TPDR), False Positive Detection Ratio (FPDR) analysis

As mentioned above, PAR is useful under a completely known information setup. In the lab setup, the location of the scatterers may be known, but it is not possible to detect which tracks hit the scatterers. Even if the track does not hit any scatterer, still we can get some deviation in the incoming and outgoing tracks due to the finite resolution of the measurement of hit point location on the detector. So a metric is certainly required to check for the degree of the true positive. True Positive Detection Ratio (TPDR) is chosen as the performance metric. TPDR is calculated considering the known location of the scatterers and is defined as the probability that a calculated PoCA point belongs to the region where scatterers are placed. Similarly, the False Positive Detection Ratio (FPDR) is the probability that it lies outside the scatterer region. Mathematically these quantities can be defined as

$$TPDR = \frac{Number\ of\ true\ positive\ PoCA\ points}{Total\ number\ of\ PoCA\ points} \tag{6.1}$$

$$FPDR = \frac{Number\ of\ false\ positive\ PoCA\ points}{Total\ number\ of\ PoCA\ points} \tag{6.2}$$

PoCA point cloud filtration algorithm aims to maximize the TPDR by removing the false positive points, by choosing the appropriate voxel size and provides a clearer reconstructed image. Figure 14 show the variation of TPDR and FPDR as a function of the volume of voxels using exact hit locations, i.e. how TPDR and FPDR vary as voxel volume decreases (or voxel dimension decrease), using PoCA point counts thresholding based algorithm. The graph indicates that using ideal data will give a pretty clear reconstructed image because the proportion of false-positive is much less, as compared to the proportion of true positives. With the simulation setup that contains 4 cubical lead blocks, the raw data itself gives the TPDR value of 0.79, which is represented by the first data point in the figure 14, that means there are only 21% false positives. These numbers change exponentially with the smeared data. These false positives can also be reduced by increasing the vertical distance between the detector planes, which results in a less fitting error and hence, better estimated hit position. Since the thresholding criteria based algorithm does not work well with smeared data, hence, standard deviation weighting factor based algorithm is used for the smeared dataset. Figure 15 shows the results of variation of TPDR, FPDR on the smeared data as a function of the volume of voxels for two different setups. The vertical distance between the detector planes is 30 cms in the first setup and 100 cm in the second setup.

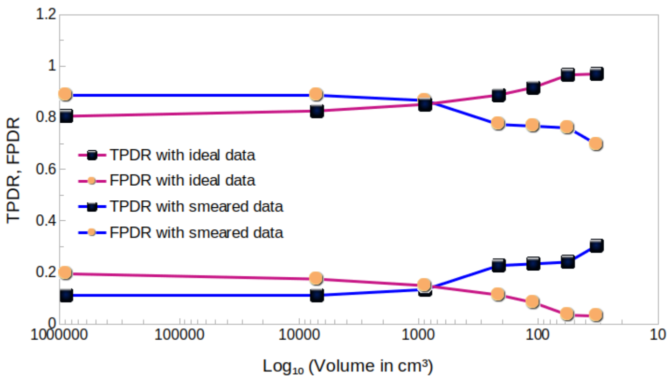

Figure 14. Plot showing variation of TPDR and FPDR on ideal and smeared simulated data for the setup with vertical distance of 30 cm between detector planes.

6.3 Analysis of the True Positives Reduction and selection of the optimal voxel size

The target of the filtration algorithm is to filter out the false positives, but the application of the filtration algorithm also removes some true positives, which affects the quality of the final reconstructed image. One cannot completely avoid the removal of true positives, but it can be controlled by choosing the proper filtration parameters. In the presented algorithms, the degree of filtration is measured by TPDR, which only depends on the number of True positives and number of false

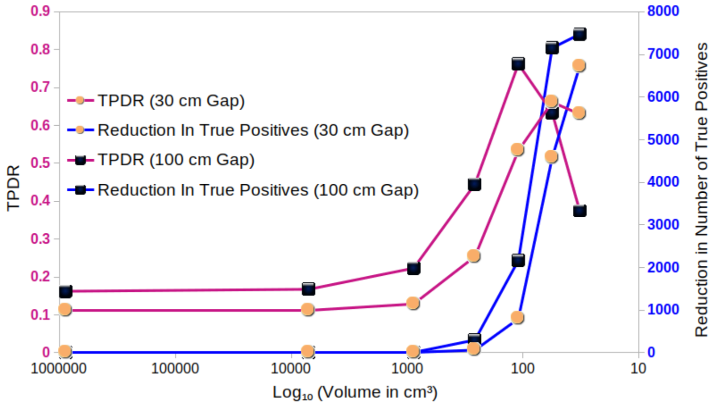

Figure 15. Plot showing variation of TPDR and true positives reduction on smeared simulated data for two different setups with vertical distance of 30 cm and 100 cm between detector planes.

positives after application of filtration algorithm, but it does not check the reduction in the number of true positives. Reduction in more true positives results in an incomplete image reconstruction and hence, requires more data to be collected, which results in increased reconstruction time. Hence, one has to carefully chose the filtration parameters. In the proposed algorithm, one critical parameter is voxel size. Hence, one has to select the optimal voxel size, to have a balance between the number of true positives and the TPDR. The selection of optimal voxel size also depends on the number of events collected and hence depends on the exposure time. It has been observed with simulated data, that increasing the number of events allows us to do the filtration with reduced voxel size. Figure 15 shows the variation of TPDR and true positive reduction on the simulated data set of 10^5 events. The volume of interest is divided into smaller voxels by dividing the side of volume to be voxelized into 5, 10, 15, 20, 25, 30 equal parts which correspond to voxels of volume 7200.00 cm^3, 900.00 cm^3, 266.66 cm^3, 112.50 cm^3, 57.60 cm^3 and 33.33 cm^3 respectively. One has to choose the proper voxel size so that the reduction in the number of true positives does not distort the final reconstructed image. From figure 15, it's clear that reducing the voxel size beyond 112.5 cm^3 results in an extremely sharp increase in the number of true positive reduction. Hence, for the used simulation setup and for 10^5 events, the optimum value of the voxel size is 112.5 cm^3, which is achieved by dividing each side of the volume enclosed between the middle two detectors in twenty equal parts. Using the selected optimal voxel size, the achieved TPDR increased from 0.16 and 0.76 for the setup where the vertical distance between detector planes is 100 cms.

7 Reconstruction results

Simulated data is used to demonstrate the result of the filtration algorithm on image reconstruction. The image reconstruction results are shown for exact hit point data, as well as for smeared data. The simulation setup is similar to, as shown in figure 10a, but the vertical distance between the RPCs is increased to 100 cm, to reduce the error due to fitting, for the case of smeared data. Top three detectors are used to get the incoming muon track, and the bottom three are used to get the outgoing muon track. Cosmic muons are generated on the top of the first detector, and 10^5 muons are used for reconstruction. Figure 16 shows the reconstruction results for ideal data, where smearing due

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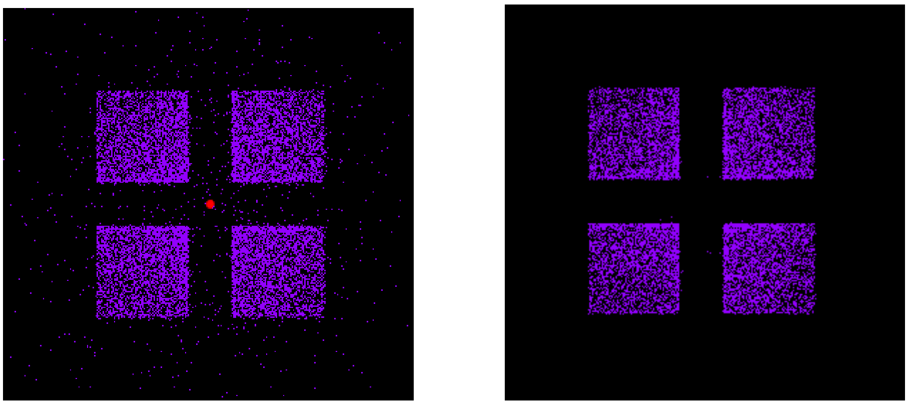
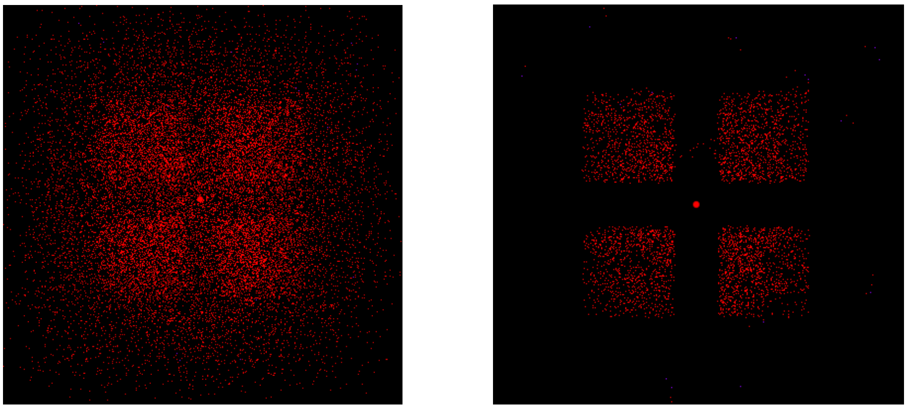
to detector resolution is not introduced. Figure 16a shows the top view of raw PoCA points. The benefit of showing the top view in the first instance is that the intensity of all the PoCA points with the same X and Y coordinate values will add up and results in a higher intensity point as compared to noise which is uniformly distributed around the objects. This results in the localization of the reconstructed object in the XY plane. The PoCA point cloud filtration algorithm is applied to raw PoCA points, using voxels of 112.5 cm^3, to remove the noisy point cloud envelope and to get the clear reconstructed image as shown in figure 16b.


(a) Top view of raw PoCA points using ideal data.

(b) Top View of filtered PoCA points using ideal data.

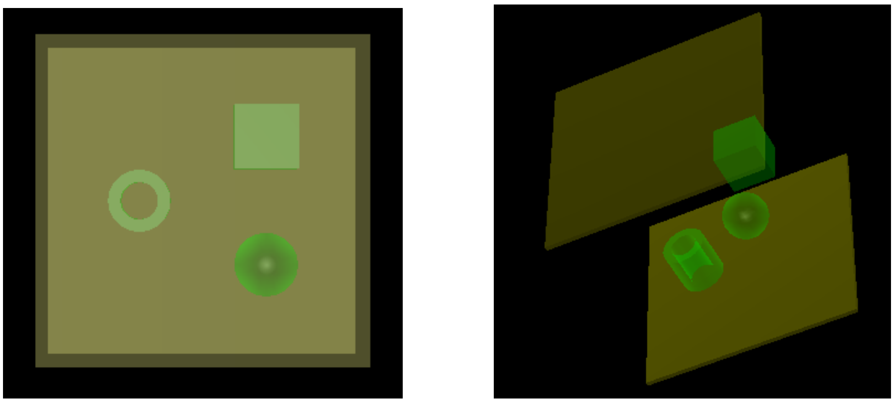
Figure 16. Reconstructed (a) Raw and (b) Filtered, PoCA points for the simulation setup shown in figure 10a, using ideal simulated data.


(a) Top view of raw PoCA points using smeared data.

(b) Top View of filtered PoCA points using smeared data.

Figure 17. Reconstructed (a) Raw and (b) Filtered, PoCA points for simulation setup shown in figure 10a, using smeared simulated data.

Figure 17 shows the reconstruction results from smeared data when the detector resolution is introduced. Figure 17a and 17b show the top view of raw and filtered PoCA points respectively, using the standard deviation (weighting factor) based algorithm. The algorithms are also tested for the scatterers of different shapes (Box, Hollow Cylinder, and Solid Sphere), as shown in the schematic in figure 18. Figure 19 shows the reconstruction results for ideal data using thresholding based criteria, which is pretty clean because there is no smearing in the data due to detector resolution. Figure 19a and 19b show the top view of raw and filtered PoCA points respectively. Figure 19c shows the perspective view of filtered PoCA points where the depth of reconstructed object can also be seen. To make sure that the PoCA points after filtration belong to the correct scatterer position, filtered PoCA points are superimposed with scatterer geometry used during the simulation. From figure 19d it is clear that filtered PoCA points lie well within the scatterer region.


(a) TopView.

(b) Perspective View.

Figure 18. Schematic showing cubical, spherical and cylindrical lead scatterers, placed between the middle two detectors of the simulation setup.

As mentioned above, thresholding criteria work very well for the exact hit point but do not work well when the data is smeared with detector resolution. The reconstruction from the smeared data is done using a standard deviation(weighting factor) based algorithm. Figure 20a and 20b show the top view of raw and filtered PoCA points respectively, using the smeared data. Figure 20c shows the perspective view of filtered PoCA points & 20d shows the perspective view of filtered PoCA points superimposed with scatterers geometry. The image quality can be improved by collecting more data.

To see the filtration capability of the algorithm on the objects of different materials, simulation is repeated with Iron as the scatterer material. The algorithm works equally well for both iron and lead and is independent of the material under test. Figure 21 and 22 below show the reconstruction using ideal simulated data and smeared simulated data respectively for the iron scatterers of different shapes.

8 Conclusion

In this paper, we have presented our work on two filtration algorithms for 3D PoCA point cloud, to get the clearer reconstructed image of the object under test. The algorithms are tested using data from the tomography setup simulated in Geant4 with the CRY library as a cosmic muon event generator. The first algorithm uses PoCA point count based thresholding to remove the false positives and work very well for exact hit point data, hence, it can be used with the detectors of fine resolution. But the experimental setup presented in the paper uses RPC with coarse resolution (~ 1 cm). This introduces smearing in the data, which results in an exponential increase in false positives,

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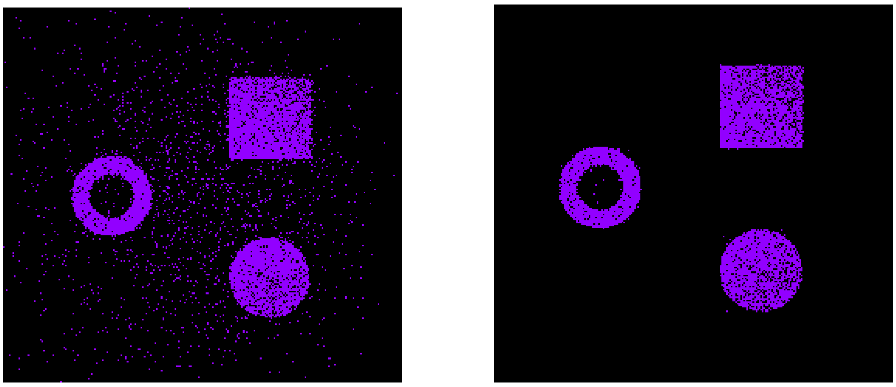
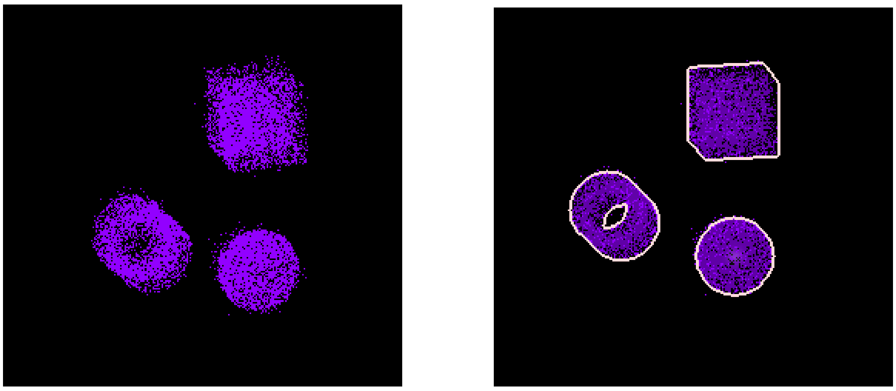


(a) Raw PoCA points using ideal data for lead scatterers of different shapes.

(b) Top view of filtered PoCA points using ideal data for lead scatterers of different shapes.

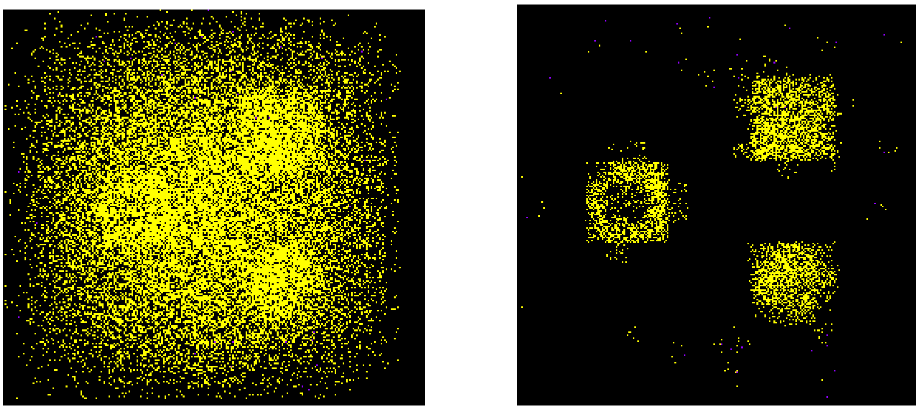


(c) Perspective view of filtered PoCA points for ideal data.

(d) Filtered PoCA points for ideal data, superimposed with scatterers geometry.

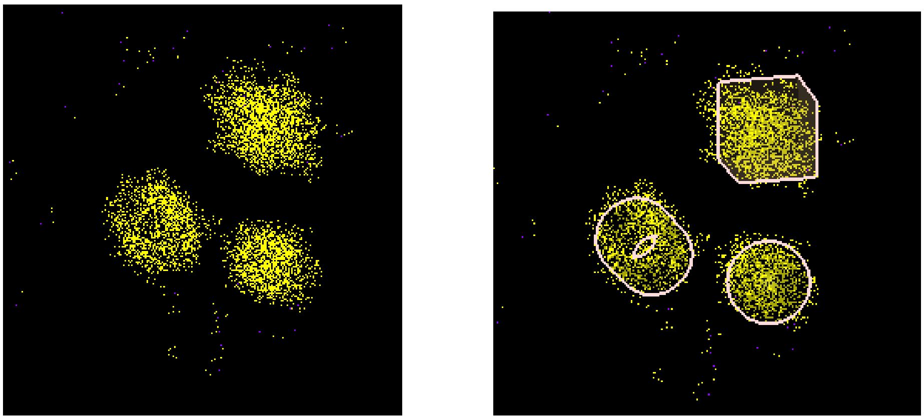
Figure 19. Reconstructed raw and filtered PoCA points using thresholding criteria, on the ideal data with lead scatterers of different shapes.

and hence, thresholding criteria on the PoCA points count in a voxel can't be used for filtration. To handle the smeared data, another algorithm is proposed, which is based on the application of the standard deviation of scattering angles of the PoCA points in voxels as a weighting factor for PoCA point counts in that voxel. This algorithm works pretty well on ideal and smeared data, and hence can be used with the experimental setup, having RPC as a position-sensitive detector. The application of the proposed algorithms have been shown on the simulation setup with scatterers of different materials and shapes. The proposed filtration algorithm has shown encouraging results for filtering out false-positive PoCA points. We will extend this work by incorporating statistical techniques to cluster the filtered PoCA points, which can eventually lead to discriminate and identify the material by calculating some material property like radiation length. We are also working on the experimental setup using RPC detectors. Two detectors are already assembled and muon tracks are reconstructed using the data from the experimental setup with these detectors. We will assemble a few more detectors to get the incoming and the outgoing muon tracks from the experimental data, to get the scattering information from the scatterers of different material and to do the improved image reconstruction using the algorithms proposed in this work.



(a) Raw PoCA points using smeared data for lead scatterers of different shapes.

(b) Top view of filtered PoCA points using smeared data for lead scatterers of different shapes.



(c) Perspective view of filtered PoCA points for smeared data.

(d) Filtered PoCA points for smeared data, superimposed with scatterers geometry.

Figure 20. Reconstructed raw and filtered PoCA points, using SD of voxel as weighting factor, on the smeared data with lead scatterers of different shapes.

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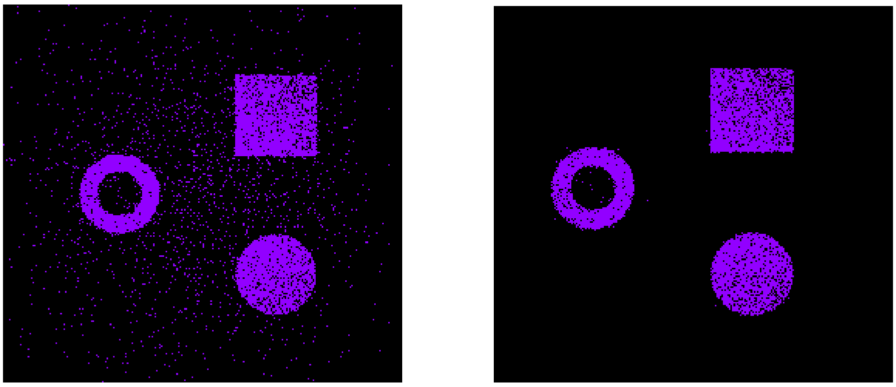
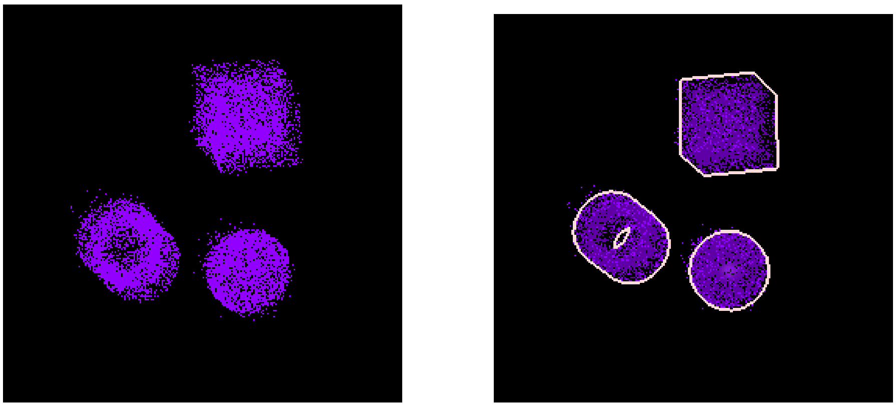


(a) Raw PoCA points using ideal data for iron scatterers of different shapes.

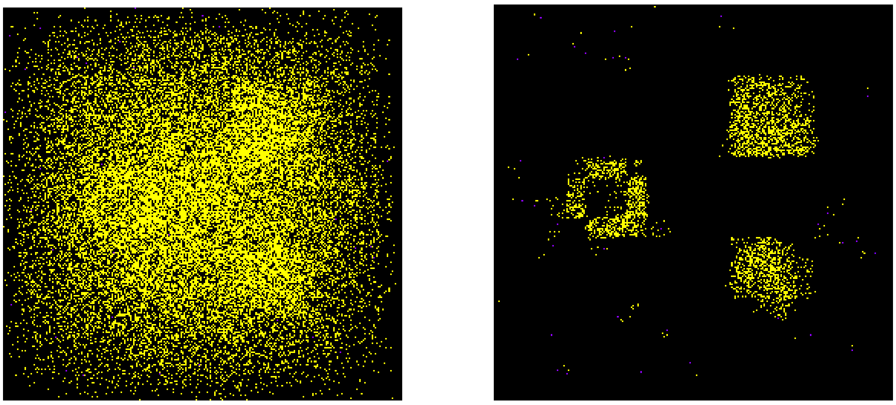
(b) Top view of filtered PoCA points using ideal data for iron scatterers of different shapes.



(c) Perspective view of filtered PoCA points for ideal data.

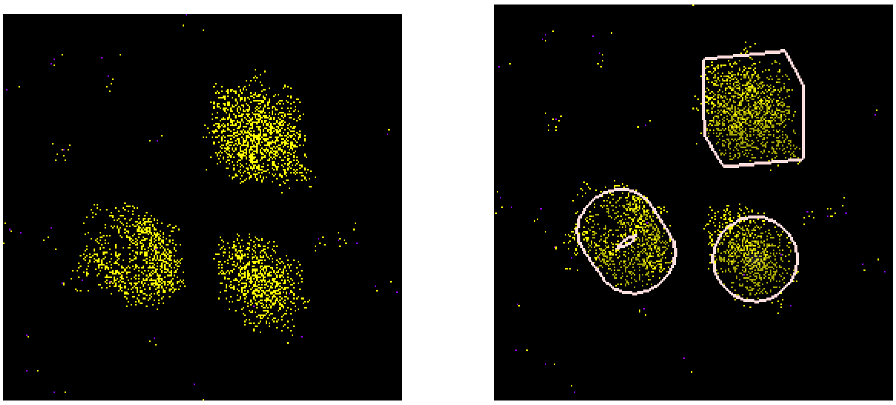
(d) Filtered PoCA points for ideal data, superimposed with scatterers geometry.

Figure 21. Reconstructed raw and filtered PoCA points using thresholding criteria, on the ideal data with iron scatterers of different shapes.



(a) Raw PoCA points using smeared data for iron scatterers of different shapes.

(b) Top view of filtered PoCA points using smeared data for iron scatterers of different shapes.



(c) Perspective view of filtered PoCA points for smeared data.

(d) Filtered PoCA points for smeared data, superimposed with scatterers geometry.

Figure 22. Reconstructed raw and filtered PoCA points, using SD of voxel as weighting factor, on the smeared data with iron scatterers of different shapes.

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